ECONOMICS AND FINANCE / 03

# Bonds and yield curves

Pricing cash flows, measuring risk



# Today's route

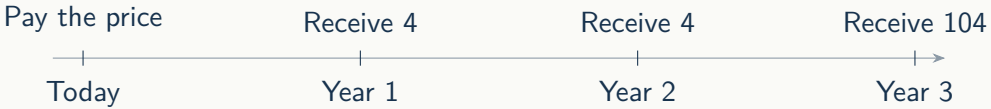
| Section | Question |
|---|---|
| Cash flows and prices | What exactly are we buying? |
| Bills and auctions | How does a new security get priced? |
| Curves and forwards | What can we lock in today? |
| Expectations and term premia | Which part of a yield is a forecast? |
| Duration and convexity | How much do prices move? |
| Hedging and market cases | What position expresses the view? |

LECTURE 3

# Cash flows and prices

What is promised, what is priced, and what can we earn?

# A bond is a dated payment contract




$$F = 100, \qquad c = 4\%, \qquad C = cF = 4.$$

# Prices and cash-flow symbols

| Symbol | Meaning |
|---|---|
| $F$ | Face value, normally 100 in our examples |
| $p_t(T)$ | Bond price at time t, with T years remaining |
| $C = cF$ | Coupon payment from coupon rate c |
| $d_t(T)$ | Price today of one certain unit paid in T years |

# Rate symbols and horizons

| Symbol | Meaning |
|---|---|
| $i_t(T)$ | Nominal spot rate for a T-year zero |
| $f_t(T_1, T_2)$ | Rate fixed today for a future lending interval |
| $i^{\mathrm{YTM}}$ | One yield matching a coupon bond's price |
| $\bar{i}_t$ | Central bank's chosen short-rate target |

$r$ still denotes a real rate. Pricing examples use nominal rates in decimals.

# A zero-coupon bond

$$p_t(T) = \frac{F}{[1 + i_t(T)]^T}.$$

$$F = 100, \quad T = 2, \quad i_t(2) = 5\% \quad \Rightarrow \quad p_t(2) = 90.70.$$

# A discount factor prices one future unit

$$d_t(T) = \frac{p_t^{\text{zero}}(T)}{F} = \frac{1}{[1 + i_t(T)]^T}.$$

$$p_t = \sum_{k=1}^{n} d_t(k)\, CF_k.$$

$CF_k$: the cash payment at date k. Match dates, currency and credit risk.

# Replication gives the pricing intuition

Buy the same cash flows using zero-coupon bonds.

A matching payoff schedule has a matching price.

Same currency, payment dates and credit quality. Borrowing, shorting and trading must be feasible.

# A coupon bond is a basket of zeros

| Date | Cash flow | Replication |
|---|---|---|
| 1 | 4 | 4 units of a date-1 zero |
| 2 | 4 | 4 units of a date-2 zero |
| 3 | 104 | 104 units of a date-3 zero |

$$p_t = 4d_t(1) + 4d_t(2) + 104d_t(3).$$

# Worked price with a flat curve

$$p_t = \frac{4}{1.05} + \frac{4}{1.05^2} + \frac{104}{1.05^3} = 97.28.$$

A 4% coupon trades below face value when the required yield is 5%.

Annual coupons. Flat 5% spot curve. Valuation on a coupon date.

# Market price determines yield to maturity

$$p_t = \sum_{k=1}^{n} \frac{CF_k}{(1 + i^{\mathrm{YTM}})^k}.$$

Given the price and promised payments, solve for the single rate.

YTM summarizes the bond price. Each dated spot rate can still be different.

# Coupon, yield and return

| Object | What does it measure? |
|---|---|
| Coupon rate c | Contractual annual payment per unit face |
| Yield $i^{\mathrm{YTM}}$ | Discount rate matching today's price |
| Holding return H | Cash received plus price change, over your horizon |

$$H_{t,t+1} = \frac{C_{t+1} + p_{t+1} - p_t}{p_t}.$$

# A 5% yield can produce a one-year loss

| Ten-year zero, face 100 | Price |
|---|---|
| Buy at 5% today | 61.39 |
| Sell one year later, remaining nine-year yield 6% | 59.19 |

$$H = \frac{59.1898 - 61.3913}{61.3913} = -3.59\%.$$

# What holding to maturity locks in

A default-free zero locks its nominal payment and annualized return.

Coupon-bond wealth also depends on coupon reinvestment.

Selling early introduces price risk. Inflation and foreign-currency returns add other risks.

# Risk-free depends on the investor's horizon

| Need | Relevant risk |
|---|---|
| 100 in ten years | A matched ten-year zero fixes that nominal payment |
| Spend after one year | A long zero has uncertain resale value |
| Roll bills for ten years | Future bill reinvestment rates are unknown |

LECTURE 3

# Bills and auctions

How does a new Treasury security get priced?

# Treasury bills are short zero-coupon claims

No periodic coupon. The investor receives face value at maturity.

| Regular original terms | Minimum purchase |
|---|---|
| 4, 6, 8, 13, 17, 26 and 52 weeks | 100 dollars |

TreasuryDirect rules checked 16 October 2026. Cash-management bills can have other terms.

# How the auction clears

Treasury announces the security, offering size and dates.

Eligible noncompetitive bids receive their requested amounts.

Competitive bids enter from low rates to high rates until the offering is allocated.

# Two ways to submit a bid

| Noncompetitive | Competitive |
| --- | --- |
| Specify the face amount. | Specify face amount and required rate. |
| Accept the auction's clearing rate. | A bid can be accepted, partly filled or rejected. |

Successful bidders pay the uniform auction price.

# An invented bill auction

| Bidder | Requested face (millions) | Discount-rate bid |
|---|---|---|
| A | 30 | 3.90% |
| B | 30 | 4.00% |
| C | 30 | 4.10% |
| D | 20 | 4.20% |

Offering: 100 million. Noncompetitive bids: 20 million. Competitive allocation remaining: 80 million.

# The cutoff bidder receives a partial fill

| Bidder | Accepted face (millions) | Outcome |
|---|---|---|
| A | 30 | Full |
| B | 30 | Full |
| C | 20 of 30 | Two-thirds |
| D | 0 | Rejected |

Stop-out discount rate $= 4.10\%$.

# One price for every successful bidder

$$p = F\left(1 - i_d \frac{N}{360}\right).$$

$$F = 100, \quad i_d = 4.10\%, \quad N = 28 \quad \Rightarrow \quad p = 99.681111.$$

$i_d$: bank-discount quotation. N: calendar days from issue to maturity.

# Actual four-week auction: 15 October 2026

| Reported item | Official result |
|---|---|
| Public offering | 110 billion dollars |
| High discount rate | 4.000% |
| Price per 100 face | 99.688889 |
| Reported bid-to-cover | 2.56 |

Issue: 20 October. Maturity: 17 November. CUSIP 912797VX2. Source: TreasuryDirect.

# The same bill has different rate quotations

$$H = \frac{F - p}{p} = 0.3121\% \quad \text{over 28 days.}$$

$$i_d = \frac{F - p}{F}\frac{360}{28} = 4.000\%.$$

$$i_{\text{inv}} = \frac{F - p}{p}\frac{365}{28} = 4.068\%.$$

Same cash flow. Different denominators and annualization rules.

# Coupon bonds also have clean and dirty prices

$$p_{\text{dirty}} = p_{\text{clean}} + \text{accrued interest.}$$

$$99.20 + 0.80 = 100.00 \quad \text{per 100 face.}$$

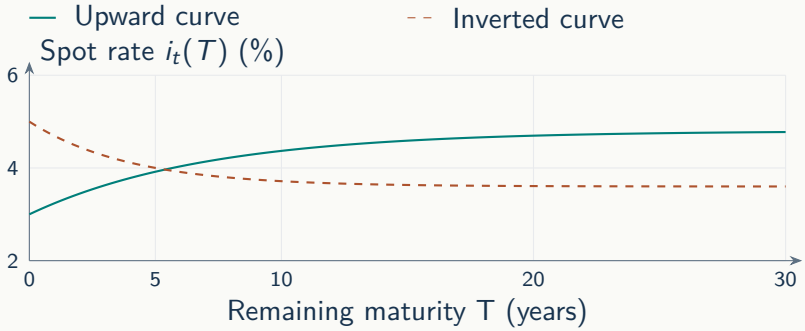
The dirty price is the cash exchanged at settlement.

LECTURE 3

# Curves and forwards

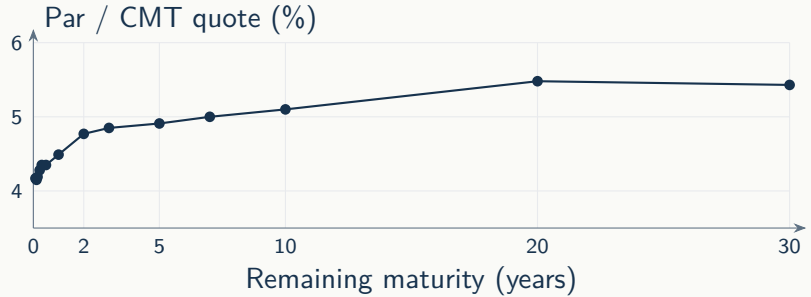
What do prices at different maturities let us lock in?

# A yield curve is one date, many maturities



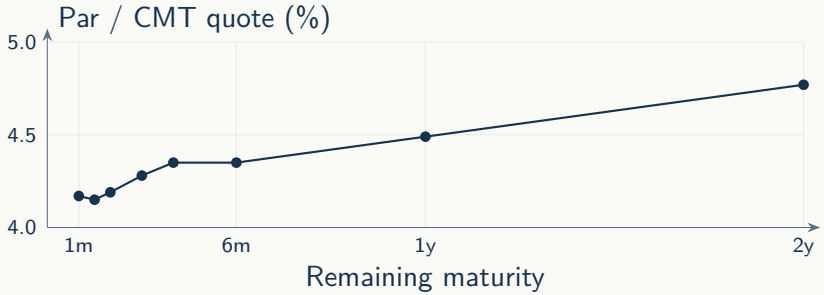

Hypothetical shapes. The horizontal axis is maturity, not the future calendar path of policy.

# US Treasury par curve: all published tenors



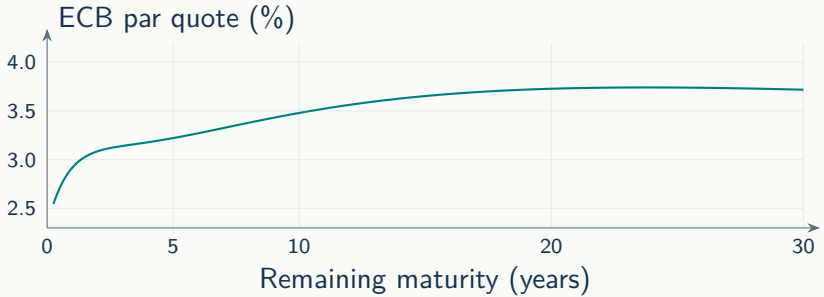

15 October 2026. All 14 published tenors, including the bill end. Treasury quotation conventions.

# The short end deserves a closer look




Same 15 October observations. A zoom changes visibility, not the data.

# Euro-area AAA: 358 fitted maturities




14 October 2026. Published monthly grid, 3 months to 30 years; fitted points are not independent quotes.

# The zero-coupon (spot) yield curve

One payment at maturity T, with no intermediate coupons.

$$i_t(T) = \left[\frac{F}{p_t^{\text{zero}}(T)}\right]^{1/T} - 1.$$

The curve plots this annualized rate against maturity T.

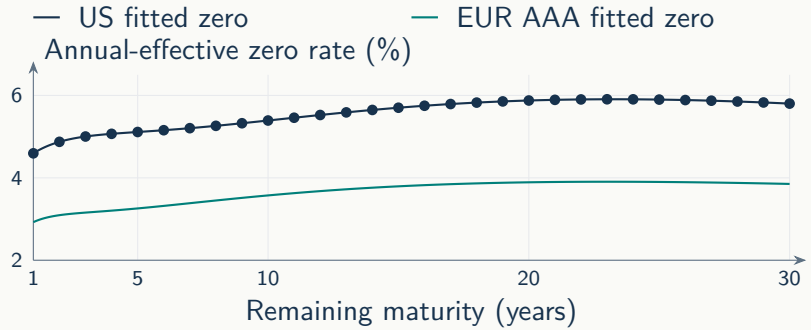
Each point discounts a single payment: $p_t^{\text{zero}}(T) = F[1 + i_t(T)]^{-T}$.

# STRIPS and estimated spot rates

| Object | Meaning |
|---|---|
| Treasury STRIP | A traded principal or coupon payment separated from a Treasury security. |
| Fitted ZCB curve | Discount factors inferred by fitting observed bond prices. |

The US panel fits off-the-run coupon bonds. Its points are model-implied ZCB yields, rather than STRIPS quotes.

# Fitted zero-coupon spot yields




9 October 2026. Continuous source yields converted by $i = \exp(z) - 1$. Different currencies and samples.

# Spot, par and YTM are different objects

| Rate | Pricing question |
| --- | --- |
| Spot $i_t(T)$ | What discounts one payment at T? |
| Par coupon rate | Which coupon makes a new bond worth F? |
| Bond-specific $i^{\mathrm{YTM}}$ | Which single rate fits this coupon schedule's price? |

More plotted maturities make the fitted shape visible. They do not add independent market quotes.

# A non-flat curve discounts dates differently

$$p_t = \frac{C}{1 + i_t(1)} + \frac{C + F}{[1 + i_t(2)]^2}.$$

Use each payment's own spot rate. The coupon rate does not choose the discount rate.

# Bootstrapping a two-year discount factor

$$d_t(1) = 0.96, \quad C = 5, \quad F = 100, \quad p_t = 100.$$

$$100 = 5(0.96) + 105 d_t(2) \quad \Rightarrow \quad d_t(2) = 0.906667.$$

$$i_t(2) = d_t(2)^{-1/2} - 1 = 5.021\%.$$

# A forward loan is agreed today




The future loan rate is fixed before the money changes hands.

# Two ways to lock a two-year payoff

| Route A | Route B |
| --- | --- |
| Buy a two-year zero. | Invest for one year. |
| Lock the whole two-year return today. | Lock year 2 using a forward agreed today. |

Same initial investment, final date, currency, credit and collateral assumptions.

# Exact no-arbitrage across the two routes

$$\boxed{[1 + i_t(2)]^2 = [1 + i_t(1)][1 + f_t(1, 2)].}$$

The forward rate adjusts until the locked final payoffs match.

# Worked forward rate

$$i_t(1) = 4\%, \qquad i_t(2) = 5\%.$$

$$f_t(1,2) = \frac{1.05^2}{1.04} - 1 = 6.0096\%.$$

$$100(1.05)^2 = 100(1.04)(1.060096) = 110.25.$$

# Replicating the forward with two zeros

| Position | Today | Year 1 | Year 2 |
|---|---|---|---|
| Short 100-face one-year zero | $+96.1538$ | $-100$ | 0 |
| Buy 106.0096-face two-year zero | $-96.1538$ | 0 | $+106.0096$ |
| Net | 0 | $-100$ | $+106.0096$ |

A future loan is a portfolio of dated claims. F is face value; f is the forward rate.

# A mispriced forward creates a locked profit

Suppose an otherwise identical quoted forward lets you lend at 8%.

Borrow synthetically at 6.0096%, then lend the same 100 at 8%.

$$\text{Year-2 net cash flow} = 108 - 106.0096 = 1.9904.$$

# A forward is a price; a forecast is a belief

| Known today | Still uncertain |
| --- | --- |
| Locked $f_t(1,2) = 6.0096\%$ | The one-year spot rate observed next year. |
| Its value follows from today's discount factors. | Its expectation needs a model or survey. |

No-arbitrage links traded prices. It does not set the physical probability of future outcomes.

LECTURE 3

# Expectations and term premia

How much of a long yield can we interpret as a forecast?

# Rolling over leaves the second rate unknown

| Year-2 rate after a 4% first year | Final wealth from 100 |
|---|---|
| 2% | $100(1.04)(1.02) = 106.08$ |
| 8% | $100(1.04)(1.08) = 112.32$ |
| Two-year zero locked at 5% | 110.25 |

An unhedged roll-over is risky. The forward-locked strategy on earlier slides is different.

# Exact compounding across maturities

$$[1 + i_t(n)]^n = \prod_{k=0}^{n-1} [1 + f_t(k, k+1)].$$

$$\ln[1 + i_t(n)] = \frac{1}{n} \sum_{k=0}^{n-1} \ln[1 + f_t(k, k+1)].$$

$f_t(0, 1) = i_t(1)$. These are rates locked today, not expected future spot rates.

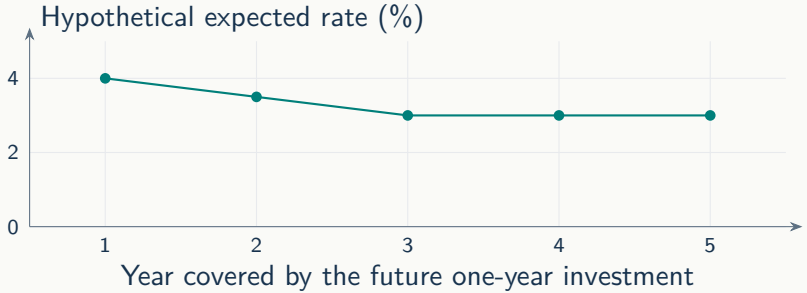
# A premium for each forward interval

$$\mu_k \equiv \mathbb{E}_t[i_{t+k}(1)], \qquad f_t(k, k+1) = \mu_k + \varphi_t(k).$$

$\varphi_t(k)$ is the forward-minus-expected-rate wedge for year k.

It can differ by horizon and observation date, and can have either sign.

# An explicit expected short-rate path




Hypothetical expectations: 4.0%, 3.5%, 3.0%, 3.0%, 3.0%. A forecast does not lock a return.

# Why a term premium can appear

A short-horizon holder of a long bond faces resale-price risk.

Investors may demand compensation for exposure to bad outcomes.

A bond that hedges the investor's bad states can instead command an insurance value.

# Premiums can differ across future years

| Year k | Expected rate | Forward premium | Locked forward |
|---|---|---|---|
| 0 | 4.0% | 0.0 points | 4.0% |
| 1 | 3.5% | 0.4 points | 3.9% |
| 2 | 3.0% | 0.6 points | 3.6% |
| 3 | 3.0% | 0.8 points | 3.8% |
| 4 | 3.0% | 1.0 points | 4.0% |

Hypothetical. In each row, $f_t(k, k+1) = \mu_k + \varphi_t(k)$.

# Compound the forward rates, period by period

$$\boxed{[1 + i_t(n)]^n = \prod_{k=0}^{n-1} [1 + \mu_k + \varphi_t(k)]\,.}$$

$$\mu_k = \mathbb{E}_t[i_{t+k}(1)], \qquad \varphi_t(k) = f_t(k, k+1) - \mu_k.$$

The premium can vary in every future period. Gross returns multiply across periods.

# Worked five-year compounded yield

$$(1 + i_t(5))^5 = 1.040 \times 1.039 \times 1.036 \times 1.038 \times 1.040.$$

$$i_t(5) = (1.040 \times 1.039 \times 1.036 \times 1.038 \times 1.040)^{1/5} - 1 = 3.8599\%.$$

With all future premiums set to zero, the same expected path gives 3.2992% under that added assumption.

# Three objects to keep separate

| Object | Whose information? | What is it? |
|---|---|---|
| Fed dots | Individual officials | Conditional policy assessments |
| Forward $f_t(T_1, T_2)$ | Market prices today | Rate on a locked future interval |
| Expectation $\mathbb{E}_t i^{(1)}$ | A model or survey | Belief about a future market rate |

# Policy cuts and rising spot yields

| Hypothetical rates | Before | After |
|---|---|---|
| Policy target $\bar{i}$ | 4.00% | 3.75% |
| Current one-year spot $i_t(1)$ | 4.00% | 3.90% |
| Year-2 expectation + premium | $3.0 + 0.5 = 3.5\%$ | $3.4 + 1.0 = 4.4\%$ |

$$i_t(2): \quad \sqrt{1.040(1.035)} - 1 \longrightarrow \sqrt{1.039(1.044)} - 1.$$

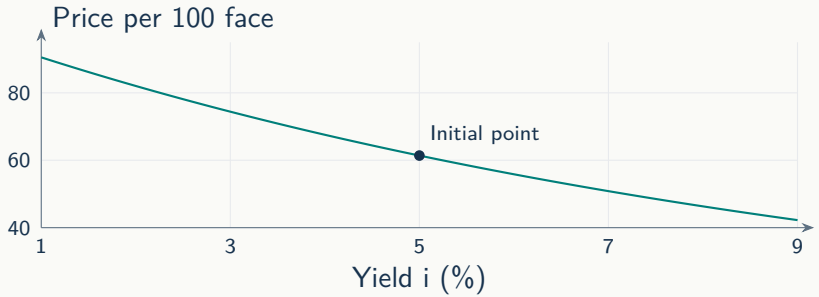
Two-year spot yield: 3.7497% to 4.1497%. The policy target and one-year spot are different rates.

LECTURE 3

# Duration and convexity

How does a discount-rate move change the price?

# Price sensitivity to a yield change




Fix cash flows and remaining maturity; vary only the yield. That yield comes from the term structure.

# Modified duration is the normalized slope

$$D_{\text{mod}} = -\frac{1}{p}\frac{dp}{di}.$$

$$\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta i.$$

A 50 bp increase means $\Delta i = 0.005$. The negative sign gives the price loss.

# A zero's duration follows by differentiation

$$p = F(1+i)^{-T}, \qquad \frac{dp}{di} = -TF(1+i)^{-T-1}.$$

$$\boxed{D_{\text{mod}} = \frac{T}{1+i}.}$$

$$T = 10, \quad i = 5\% \quad \Rightarrow \quad D_{\text{mod}} = 9.5238.$$

# Coupon duration weights the payment dates

| Date $k$ | Cash flow | PV weight |
|---|---|---|
| 1 | 4 | 3.92% |
| 2 | 4 | 3.73% |
| 3 | 104 | 92.35% |

$$D_{\text{Mac}} = \sum_k k\, w_k = 2.8844 \text{ years.}$$

# Macaulay and modified duration

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+i}.$$

$$D_{mMac} = 2.8844 \quad \Rightarrow \quad D_{\text{mod}} = 2.7470 \quad \text{at } i = 5\%.$$

Annual coupon convention. Payment timing gives Macaulay duration; the derivative gives modified duration.

# Dollar value of one basis point: DV01

DV01 means the dollar value of a one-basis-point yield move.

$$DV01 \approx VD_{\text{mod}} \times 10^{-4}.$$

$$V = \$1{,}000{,}000, \quad D_{\text{mod}} = 8 \quad \Rightarrow \quad DV01 = \$800/\text{bp}.$$

One bp $= 0.01$ percentage point $= 0.0001$. A 1 bp rise costs a long position about \$800.

# A ten-basis-point move

$$\Delta i = +10 \text{ bp} \quad \Rightarrow \quad \Delta V \approx -800 \times 10 = -8{,}000.$$
$$\Delta i = -10 \text{ bp} \quad \Rightarrow \quad \Delta V \approx +8{,}000.$$

The linear approximation is symmetric. The exact price curve is curved.

# Duration comes from a Taylor expansion

$$p(i + \Delta i) = p(i) + p'(i)\Delta i + \frac{1}{2}p''(i)(\Delta i)^2 + O((\Delta i)^3).$$

The first derivative is the tangent. The second derivative measures curvature.

Cash flows and maturity are fixed while the yield parameter changes.

# Convexity normalizes the second derivative

$$\mathcal{C} = \frac{1}{p}\frac{d^2 p}{di^2}.$$

$$\boxed{\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta i + \frac{1}{2}\mathcal{C}(\Delta i)^2.}$$

For a noncallable bond with fixed positive payments, the price–yield convexity is positive.

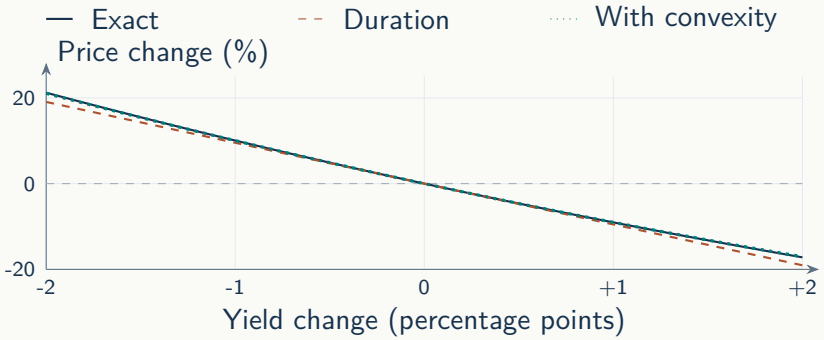
# A zero's convexity is explicit

$$\frac{d^2p}{di^2} = T(T+1)F(1+i)^{-T-2}.$$

$$\boxed{\mathcal{C} = \frac{T(T+1)}{(1+i)^2}.}$$

$$T = 10, \quad i = 5\% \quad \Rightarrow \quad \mathcal{C} = 99.7732.$$

# Exact, tangent and quadratic repricing




Ten-year zero, initial yield 5%, face 100. The quadratic follows the exact curve more closely locally.

# Work out a one-percentage-point move

| Yield move | Exact | Duration | With convexity |
|---|---|---|---|
| -1 point | +10.042% | +9.524% | +10.023% |
| +1 point | -9.043% | -9.524% | -9.025% |

$$\tfrac{1}{2}\mathcal{C}(0.01)^2 = 0.0049887 \approx 0.499\%.$$

# The asymmetry has economic meaning

For equal rate moves, the gain from a fall exceeds the loss from a rise.

More distant fixed payments generally amplify the curvature.

Positive convexity is valuable, but securities with different convexity need not have the same price or yield.

# Sensitivity is conditional on the experiment

A single-yield bump isolates one price-risk channel.

Curve twists, credit spreads and changing cash flows need additional exposures.

Duration matching removes a local parallel-move exposure. It does not make a portfolio riskless.

# A liability changes what counts as risky

| Ten-year nominal liability of 100 | Investment choice |
|---|---|
| Match the cash flow | A ten-year default-free zero pays 100 on the required date |
| Roll short bills | The future reinvestment path is uncertain |
| Trade a long bond over one year | The sale price is uncertain |

# A bond fund usually keeps exposure beyond one maturity

A single bond has a contractual maturity payment.

A rolling bond fund continually replaces holdings and retains a duration exposure.

The relevant comparison is the investor's cash-flow need and holding horizon.

LECTURE 3

# Hedging and market cases

How does a finance view become a position with a measurable risk?

# News lab: income horizon or price trade?

A WSJ bond-strategy column asks investors to identify their objective.

Compare a fixed spending need with a forecast of a bond-price rally.

Telis Demos, WSJ, 8 October 2026. Discussion prompt, not a trade recommendation.

Read the WSJ column

# News lab: buying during a Treasury sell-off

FT reporting describes investors adding Treasuries after a substantial rise in yields.

A higher starting yield improves the quoted entry rate; a further rate rise can still create a near-term loss.

Clarfelt and Duguid, Financial Times, 13 October 2026. Separate the report from our hypothetical exercise.

Read the Financial Times article

# Two real auctions, two different questions

| Auction | Offering | High yield | Bid-to-cover |
|---|---|---|---|
| 7 Oct: ten-year note reopening | 39 bn | 5.300% | 2.77 |
| 8 Oct: thirty-year bond reopening | 22 bn | 5.618% | 2.54 |

TreasuryDirect results, October 2026. These are auction yields, not coupon rates or total returns.

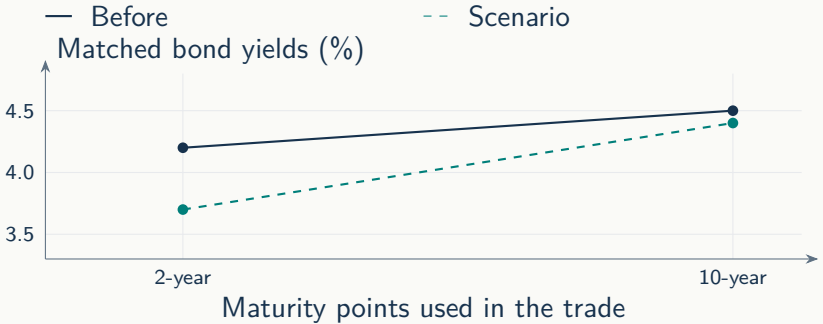
# An auction tail needs a market benchmark

$$\text{Tail} = \text{auction stop yield} - \text{when-issued yield at the auction deadline.}$$

$$5.30\% - 5.28\% = 0.02 \text{ points} = 2 \text{ bp.}$$

Illustrative numbers. Same issue, timestamp and quote convention are required.

# A curve view needs two maturity scenarios




Two risk points in an invented trade scenario. This is not a sparse empirical curve estimate.

# This scenario is a bull steepening

$$\text{Slope} = i_{10} - i_2.$$

$$4.50\% - 4.20\% = 0.30\% \quad \longrightarrow \quad 4.40\% - 3.70\% = 0.70\%.$$

Yields fall. The long-minus-short yield spread widens.

# Expressing the relative view

| Leg | Benefit | Risk in this scenario |
|---|---|---|
| Buy two-year bond | Its yield falls 50 bp | Positive price contribution |
| Short ten-year bond | Its yield rises | Its 10 bp fall produces a loss |

A slope view needs a risk hedge. Equal face amounts do not match rate sensitivity.

# Match DV01, rather than market values

| Position | Market value | Modified duration | DV01 |
|---|---|---|---|
| Long two-year | 1,000,000 | 1.8 | 180 / bp |
| Short ten-year | ? | 8.0 | 180 / bp |

$$V_{\text{short}} = \frac{180}{8(0.0001)} = 225{,}000.$$

# Work out the scenario profit and loss

| Leg | Yield move | Approximate P&L |
|---|---|---|
| Long two-year | $-50$ bp | $+180(50) = +9{,}000$ |
| Short ten-year | $-10$ bp | $-180(10) = -1{,}800$ |
| Combined | 40 bp steepening | $+7{,}200$ |

First-order repricing before carry, financing, trading costs and convexity.

# A small parallel move cancels at first order

$$\Delta i_2 = \Delta i_{10} = +25 \text{ bp.}$$

$$\Delta V \approx -180(25) + 180(25) = 0.$$

The spread does not change. Higher-order risk and financing remain.

# Counterfactual: the curve flattens

| Leg | Alternative yield move | Approximate P&L |
|---|---|---|
| Long two-year | $-10$ bp | $+1{,}800$ |
| Short ten-year | $-50$ bp | $-9{,}000$ |
| Combined | 40 bp flattening | $-7{,}200$ |

Both bonds rally, but this steepener loses.

# Price impact of the policy-cut scenario

$$\Delta i_2 \approx +40 \text{ bp} = +0.004.$$

$$V = 1{,}000{,}000, \quad D_{\text{mod}} = 1.8 \quad \Rightarrow \quad \Delta V \approx -7{,}200.$$

$$\mathcal{C} = 5 \quad \Rightarrow \quad \tfrac{1}{2} V \mathcal{C} (0.004)^2 = +40.$$

Approximate repricing: $-7{,}160$. The yield shock is rounded; carry and financing are excluded.

# Questions before taking the trade

Which future cash flow, rate shock and holding horizon are being priced?

How does the forecast differ from the forward curve or market baseline?

What happens under the opposite curve move, after financing and costs?

LECTURE 3

# Optional foundations

How do dates, states and compounding fit one pricing framework?

# General forward intervals

$$[1 + f_t(T_1, T_2)]^{T_2 - T_1} = \frac{d_t(T_1)}{d_t(T_2)}.$$

$$f_t(T_1, T_2) = \left[\frac{d_t(T_1)}{d_t(T_2)}\right]^{1/(T_2 - T_1)} - 1.$$

Annual-effective forward convention. Actual contracts can use simple interest and specific day counts.

# Continuous and annual-effective spot rates

$$d_t(T) = e^{-z_t(T)T} = [1 + i_t(T)]^{-T}.$$

$$i_t(T) = e^{z_t(T)} - 1.$$

The Fed and ECB zero inputs use continuous rates z. Our matched-date comparison converts them to i.

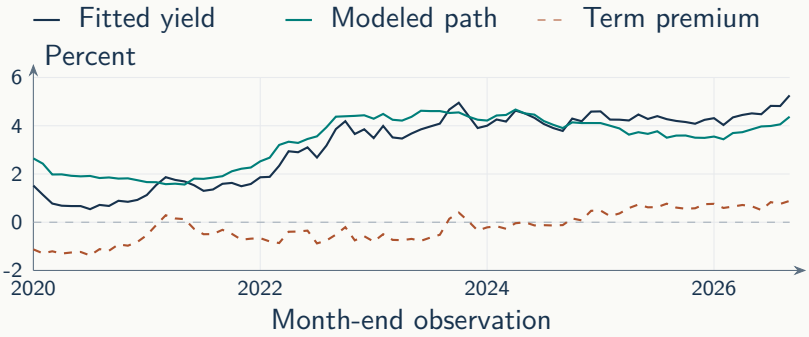
# The par coupon comes from discount factors

$$F = c_{\mathrm{par}} F \sum_{k=1}^{n} d_t(k) + F d_t(n).$$

$$\boxed{c_{\mathrm{par}} = \frac{1 - d_t(n)}{\sum_{k=1}^{n} d_t(k)}.}$$

Annual coupon dates and unit year fractions in this simplified formula.

# Optional: ACM estimates




NY Fed ACM model, ten-year maturity. January 2020–September 2026. Components are estimates.

# ACM components use a model convention

$$4.37\% + 0.89\% = 5.26\% \quad \text{approximately.}$$

| 30 September 2026 | ACM estimate |
|---|---|
| Modeled rate-path component | 4.3740% |
| Term premium | 0.8850% |
| Fitted ten-year yield | 5.2590% |

An additive reporting convention of this model. The core pricing relation compounds forwards.

# Two states: equal average payoff, different price

| Payoff | Good state | Bad state | Expected payoff | Price |
|---|---|---|---|---|
| A | 100 | 0 | 50 | 40.00 |
| B | 0 | 100 | 50 | 56.15 |

Original two-state example. Both states have probability 1/2. Payments in the bad state are more valuable.

# A stochastic discount factor prices states

$$p_t = \mathbb{E}_t[m_{t+1} X_{t+1}].$$

$$p_t = \mathbb{E}_t[m]\,\mathbb{E}_t[X] + \mathrm{Cov}_t(m, X).$$

X: future payoff. m: value today per unit paid in that state, adjusted for its probability.

# Optional liability hedge: match value and duration

Liability: 100 in ten years, $i = 5\%$.

$$w_1 + w_{20} = 1, \qquad 1w_1 + 20w_{20} = 10.$$

$$w_1 = \frac{10}{19}, \qquad w_{20} = \frac{9}{19}.$$

Weights are current market-value shares in one- and twenty-year zeros.

LECTURE 3

# Reading

Where can we deepen the finance and market perspective?

# Reading and next questions

Bodie, Kane and Marcus: pricing, term structure and bond-risk management.

The two supplied finance courses: replication, realized returns, auctions and immunization.

Fed, Treasury and ECB sources; WSJ and FT discussion readings via MIT.

Exact chapters, page mappings, dates and links are in the speaker notes and reading list.